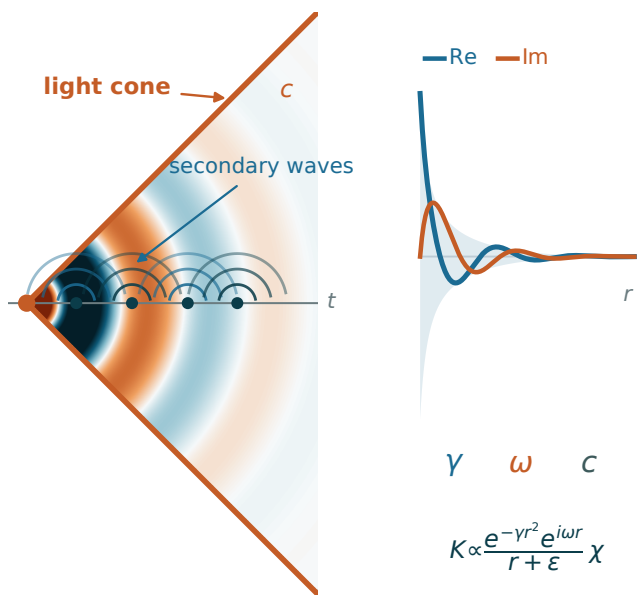
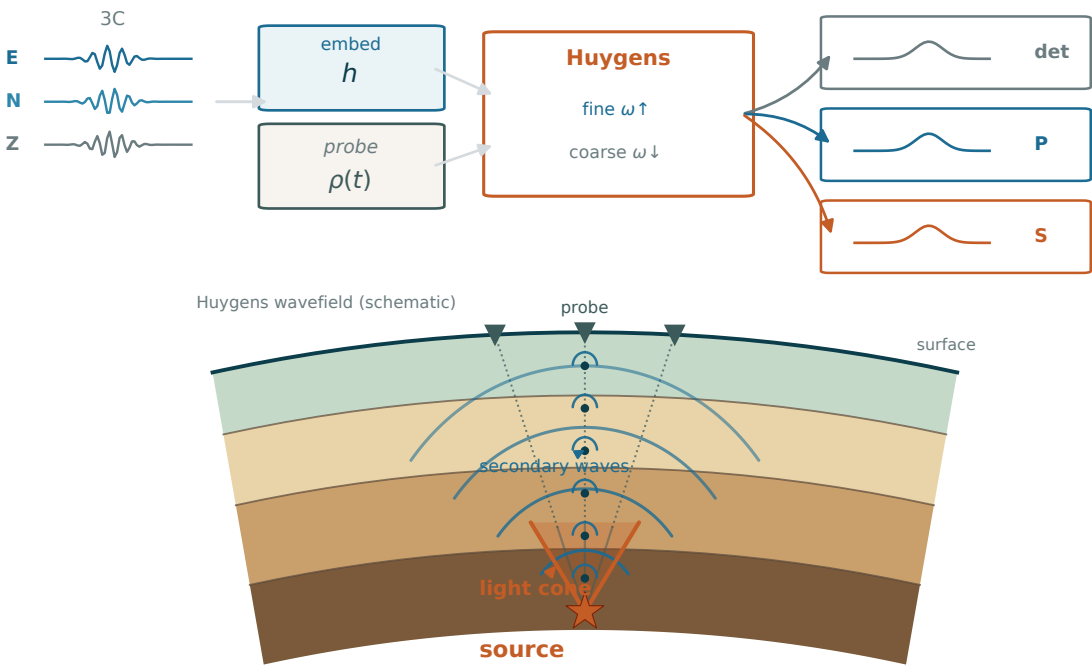


Physics Neural Field (PNF): principle, latents, and discovery paradigm

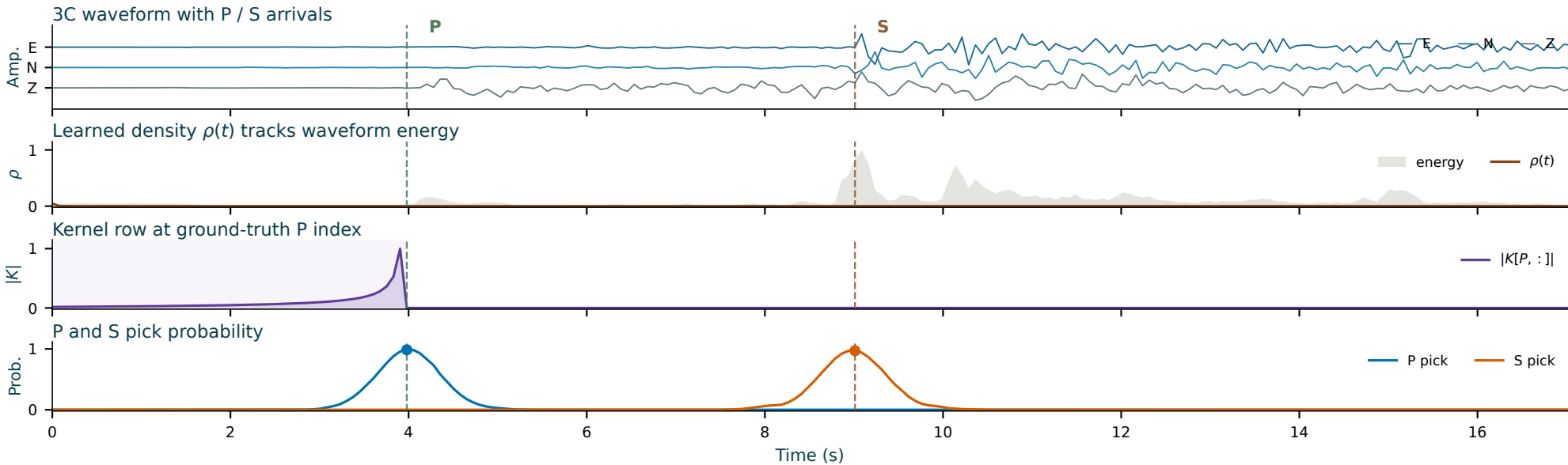
a Huygens principle → learnable kernel



b Forward path: STEAD 3C picking



c Interpretable latent variables in the seismic domain

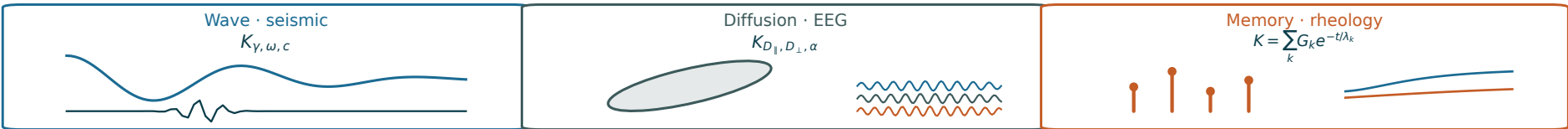


d PNF: causal physics kernel for new discoveries

Causal integral operator

$u = \mathcal{K}_\theta[s] = \int_{\tau \leq t} K_\theta(t - \tau) s(\tau) d\tau$ · light-cone support · θ readable

Physics kernel family



Workflow for discoveries

